## EC22-3: Power BI

| Teaching Scheme: | Credit: 03 | Examination Scheme: |
|---|---|---|
| **Theory Session**: 45 Hours | | **Internal(TH)**: 25 Marks<br>**External (TH)** : 50 Marks<br>**Total :75 Marks** |

**Prerequisites:** Database Knowledge, Business Understanding

**Course Objectives:**

- To utilize Power BI tools effectively for data connectivity, transformation, and visualization.
- To Apply data modelling techniques to build relationships and optimize data analysis.
- To Incorporate slicers, filters, and bookmarks to enhance user interactivity and exploration.
- To Understand Power BI concepts like Microsoft Power BI desktop layouts, BI reports, dashboards, and Power BI DAX commands and functions
- To Gain a competitive edge in creating customized visuals and deliver a reliable analysis of vast amount of data using Power BI

**Course Outcomes:**

On completion of the course, learners should be able to

| CO# | Cognitive Domain | Course Outcomes |
|---|---|---|
| CO1 | Apply | Demonstrate the concepts and importance of data modelling, data source, data cleaning, data transformation in Power BI. |
| CO2 | Analyse | Analyse data relationships and model data using DAX |
| CO3 | Analyse | Assess the interactivity of visualizations using slicers, filters, and drill through features. |
| CO4 | Apply | Use M Queries to extract, transform, and load data from various sources |
| CO5 | Analyse | Examine Power BI solutions that solve real-world business problems as outlined in case studies |

| Unit No. | Contents | Weightage in % | No of Sessions |
|---|---|---|---|
| 1 | **Introduction to Data Visualization and BI**<br>1.1 Overview of Business Intelligence (BI)<br>1.2 Introduction to Power BI<br>1.3 Data Modelling in Power BI<br>1.3.1 Introduction to data modelling concepts<br>1.3.2 Creating and managing relationships between tables<br>1.3.3 Star schema and snowflake schema<br>1.3.4 Data normalization and de-normalization<br>1.4 Data Visualization Tools<br>1.4.1 Power BI<br>1.4.2 Tableau | 15 | 8 |

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|---|---|---|---|
| | 1.4.3 Google Data Studio<br>1.4.4 Microsoft Excel<br>1.5 Power BI Desktop and Data Transformation<br>1.5.1 Overview of Data Preparation<br>1.5.2 Data Connection and Import<br>1.5.2.1 Connecting to Different Data Sources<br>1.5.2.2 Direct Query vs. Import Mode<br>1.5.3 Data Cleaning Basics<br>1.5.3.1 Handling Missing Data<br>1.5.3.2 Data Deduplication<br>1.5.3.3 Handling Outliers<br>1.5.4 Data Transformation Technique<br>1.5.4.1 Merging and Appending Queries<br>1.5.4.2 Pivoting and Unpivoting Data<br>1.5.4.3 Using Conditional Columns<br>1.5.5 Data Formatting and Structuring<br>1.5.5.1 Data Formatting<br>1.5.5.2 Creating Custom Columns<br>1.5.5.3 Grouping and Aggregating Data | | |
| *Mapping of Course Outcomes for Unit 1: CO1 | | | |
| 2 | **Filter and Data Analysis Expression (DAX)**<br>2.1 Filtering Data<br>Using Slicers, Visual Filters, Page Filters, Report Level, Drill Through Filter, cross report filters<br>2.2 DAX in Power BI<br>2.2.1 Introduction of DAX<br>2.2.2 Data Types in DAX<br>2.2.3 DAX Formula – Syntax<br>2.2.4 DAX Calculation Types<br>2.2.5 Steps to Create Calculated Columns<br>2.2.6 Measures in DAX<br>2.2.7 DAX Functions<br>2.2.8 DAX Operators<br>2.2.9 DAX Tables and Filtering | 15 | 7 |
| *Mapping of Course Outcomes for Unit 2:CO2 | | | |
| 3 | **Data Visualization and Reports**<br>3.1 Types of Report<br>3.1.1 Standard Reports<br>3.1.2 Interactive Reports<br>3.1.3 Paginated Reports<br>3.1.4 Dashboards<br>3.1.5 Analytical Reports<br>3.1.6 Custom Reports | 20 | 10 |

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| | 3.2 Visualization<br>3.2.1 Visualization Charts in Power BI<br>3.2.2 Matrixes and Tables<br>3.2.3 Slicers and Map Visualizations<br>3.2.4 Gauges and Single Number Cards<br>3.2.5 Modifying Colors in Charts and Visuals Shapes, Text Boxes, and Images<br>3.2.6 Custom Visuals<br>3.2.7 Page Layout and Formatting<br>3.2.8 Bookmarks and Selection Pane<br>3.2.9 KPI Visuals<br>3.2.10 Z-order<br>3.2.11 Grouping and Binding<br>3.3 Introduction to Power BI Service<br>3.3.1 Creating a Dashboard<br>3.3.2 Quick Insights in Power BI<br>3.3.3 Configuring a Dashboard<br>3.3.4 Power BI Q&A<br>3.3.5 Ask Questions about your Data<br>3.3.6 Power BI Embedded<br>3.3.7 Bookmarks and buttons | | |
| *Mapping of Course Outcomes for Unit 3:CO3 | | | |
| 4 | **Introduction of SQL Server**<br>4.1 Power Query & M Language<br>4.1.1 Introduction to Power Query and M Language<br>4.1.2 Introduction to Power Query Editor<br>4.1.3 Understanding M language fundamentals<br>4.1.4 Basic M Query syntax and functions<br>4.1.5 Data types and operators in M Query<br>4.2 Data Transformation with M Query<br>4.2.1 Importing and cleaning data<br>4.2.2 Filtering, sorting, and grouping data<br>4.2.3 Pivoting and unpivoting columns<br>4.2.4 Merging and appending queries<br>4.2.5 Creating custom functions<br>4.2.6 Error handling in M Query | 25 | 10 |
| *Mapping of Course Outcomes for Unit 4:CO4 | | | |
| 5 | **Real World Use Cases and Case studies**<br>5.1 Real-World Use Cases<br>5.1.1 Financial Services-Risk Management<br>5.1.2 Healthcare-Patient Care Improvement<br>5.1.3 Retail-Sales Performance Analysis<br>5.1.4 Education-Student Performance Monitoring | 25 | 10 |

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|---|---|---|---|
| | 5.1.5 Manufacturing-Production Line Optimization<br>5.1.6 Marketing-Campaign Performance Analysis<br>5.2 Case Studies<br>Charles Schwab, The Texas Rangers, Deloitte, University of British Columbia, Cisco, Tata Consultancy Services (TCS), ICICI Bank, Reliance Industries Limited (RIL), Flipkart, Indian School of Business (ISB) | | |
| *Mapping of Course Outcomes for Unit 5:CO5 | | | |

**Learning Resources**

**Text Books:**

- Mastering Microsoft Power BI" by Brett Powell
- "Analyzing Data with Power BI and Power Pivot for Excel" by Alberto Ferrari and Marco Russo
- "Microsoft Power BI Cookbook: Creating Business Intelligence Solutions of Analytical Data Models, Reports, and Dashboards" by Brett Powell -

**Reference Books:**

- Business Intelligence Guidebook: From Data Integration to Analytics" by Rick Sherman
- "Pro Power BI Desktop" by Adam Aspin
- "The Definitive Guide to DAX, Second Edition: Business intelligence with Microsoft Excel, SQL Server Analysis Services, and Power BI" by Marco Russo and Alberto Ferrari
- "Successful Business Intelligence: Unlock the Value of BI & Big Data" by Cindi Howson
- "Mastering Microsoft Power BI: Expert techniques for effective data analytics and business intelligence" by Brett Powell

**Recommended Learning Material:**

- Microsoft Learn: Power BI Learning Path
- https://docs.microsoft.com/en-us/learn/powerplatform/power-bi
- Microsoft Learn: Introduction to DAX in Power BI
- https://docs.microsoft.com/en-us/learn/modules/dax-power-bi/
- Power BI Documentation - Microsoft Docs
- https://docs.microsoft.com/en-us/power-bi/

**Recommended Certification:**

- LinkedIn Learning: Learning Power BI
- Udemy: Power BI A-Z: Hands-On Power BI Training for Data Science!
- Coursera: Data Visualization with Power BI Specialization

## EC22-4: Essentials of Information Security

| Teaching Scheme: | Credit: 03 | Examination Scheme: |
|---|---|---|
| **Theory Sessions**: Total 45 Hours | | **Internal (TH)**: 25 Marks<br>**External (TH)**: 50 Marks<br>**Total :75 Marks** |

**Prerequisites:** Basic knowledge of Cyber Security

**Course Objectives:**

- Conduct a cyber security risk assessment using tool.
- Measure the performance and troubleshoot audit.
- Design and develop a security architecture for an organization.
- Design operational and strategic cyber security strategies and policies.

**Course Outcomes:**

On completion of the course, learners should be able to

| CO# | Cognitive Domain | Course Outcomes |
|---|---|---|
| CO1 | Understand | Understand the fundamental concepts of cybersecurity, including its importance and various threats in cyberspace. |
| CO2 | Understand | Understand the vulnerable to threats in systems |
| CO3 | Apply | Design and Apply the need for security architecture and its relevance to systems, service continuity and reliability |
| CO4 | Understand | Ability to describe the various auditing tools that can be used in cybersecurity management |
| CO5 | Apply | Identifies the needs of users in the field of developing information systems and building secure computer networks. |

| Unit No. | Contents | Weightage in % | No of Sessions |
|---|---|---|---|
| 1 | **CYBER SECURITY ESSENTIALS**<br>1.1 Information Assurance Fundamentals<br>1.1.1 Basic Cryptography<br>1.1.2 Symmetric Encryption<br>1.1.3 Public Key Encryption<br>1.1.4 The Domain Name System (DNS)<br>1.1.5 Firewalls<br>1.1.6 Virtualization<br>1.1.7 Radio-Frequency Identification<br>1.2 Microsoft Windows Security Principles<br>1.2.1 Windows Tokens<br>1.2.2 Window Messaging<br>1.2.3 Windows Program Execution<br>1.2.4 The Windows Firewall | 20 | 9 |
| *Mapping of Course Outcomes for Unit 1: CO1 | | | |
| 2 | **Information Security** | 15 | 6 |

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|---|---|---|---|
| | 2.1 Introduction<br>2.2 Security Threat Supply<br>2.3 Information Assurance<br>2.4 Quantitative Risk Analysis Techniques and Tools<br>2.5 Introduction to IT Auditing and Reporting Techniques | | |
| *Mapping of Course Outcomes for Unit 2: CO2, CO4 | | | |
| 3 | **Development of Secure Information System**<br>3.1 Introduction<br>3.2 Developing Secure Information Systems<br>3.3 Key Elements of an Information Security Policy<br>3.4 Information System Development Life Cycle<br>3.5 Application Security<br>3.6 Information Security Governance<br>3.8 Security Architecture and Design<br>3.9 Case Study based information system design | 25 | 12 |
| *Mapping of Course Outcomes for Unit c 3: CO3 | | | |
| 4 | **Security Threats and Policies**<br>4.1 Introduction to Security Threats<br>4.2 Network and Denial of Services Attack<br>4.3 Security Threats to E-Commerce<br>4.4 Introduction to Security Policies<br>4.5 Why can we would like Security Policy?<br>4.6 Security Policy Development<br>4.7 Email Security Policies<br>4.8 Advanced persistent threat<br>4.9 Case Study based on security threat and policy | 25 | 12 |
| *Mapping of Course Outcomes for Unit 4: CO4 | | | |
| 5 | **Securities in Operating System And Networks**<br>5.1 Introduction to Securities in Operating System Network<br>5.2 Rootkit and Anti Rootkit Tools (Antivirus Based)<br>5.3 Threats to Network Communication<br>5.4 Wireless Network Security<br>5.5 Network Security Attack | 15 | 6 |
| *Mapping of Course Outcomes for Unit 5: CO5 | | | |
| **Learning Resources** | | | |

**Text Books**

- Michael E. Whitman, Herbert J. Mattord, (2018). Principles of Information Security, 6th edition, Cenage Learning, N. Delhi
- Cryptography and Network Security by William Stallings
- Network Security Essentials by William Stallings
- Computer Security and the Internet: Tools and Jewels from Malware to Bitcoin,

Second Edition, by Paul C. van Oorschot. Springer, 2021.
- Applied Cryptography by Bruce Schneier

**Reference Books**

- Computer Security: Principles and Practice by Stallings and Brown
- Computer Security by Dieter Gollmann
- Information Security: Principles and Practice (2011, 2/e; Wiley) by Mark Stamp
- Hacking: The Art of Exploitation by Jon Erickson
- The Web Application Hacker's Handbook by Dafydd Stuttard and Marcus Pinto
- Web Security Sourcebook: A Complete Guide to Web Security Threats and Solutions by Rubin, Geer and Ranum
- Cybersecurity Essentials" by Charles J. Brooks, Christopher Grow, Philip Craig, and Donald Short
- "Introduction to Cyber Security: Stay Safe Online" by Simplilearn
- Information Security Policies, Procedures, and Standards: Guidelines for Effective Information Security Management" by Thomas R. Peltier

**Recommended Learning Material**

- www.unodc.org
- www.studocu.com
- cod.pressbooks.pub

**Recommended Certification**

- Certificate in Information Systems Audit and Control Association (ISACA)
- Certified Information Systems Security Professional (CISSP)
- Certified Information Security Manager (CISM)
- Certified Information Systems Auditor (CISA)
- Certified Information Privacy Professional (CIPP)
- Certified Information Security Manager (CISM)